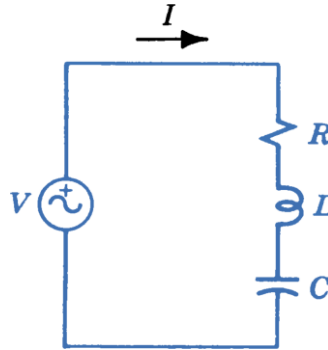


Tutorial 6

ELL-101

1. An industrial plant consists of the following loads: (i) 1 heating load of 5 kW and (ii) one motor load of 50 kVA at 0.6 lagging power factor. Power is supplied at 240 Volts.
 - a. Find the total current magnitude that needs to be supplied to this plant.
 - b. Find the plant power factor.
 - c. You propose to improve the power factor of the plant to 0.95 lagging by connecting auxiliary equipment. Find the specifications of that equipment.
 - d. With the improved power factor, find the total current magnitude that needs to be supplied to this plant.
 - e. By observing part (a) and (d), comment whether your proposal to improve the plant power factor was worth it or not.
2. A series circuit consists of a $0.5\ \mu\text{F}$ capacitor, a coil of inductance $0.32\ \text{H}$ and resistance $40\ \Omega$, and a $20\ \Omega$ non-inductive resistor. Calculate the resonant frequency of the circuit. When the circuit is connected to a 30 V AC supply at this resonant frequency, determine:
 - (a) The potential difference across each of the three components.
 - (b) The current flowing in the circuit.
 - (c) The active power absorbed by the circuit.
3. A circuit consists of a $10\ \Omega$ resistor, a $30\ \text{mH}$ inductor, and a $1\ \mu\text{F}$ capacitor. It is supplied from a 10 V variable-frequency source. Find the frequency at which the voltage developed across the capacitor is maximum and calculate the magnitude of this voltage.
4. The variable capacitor behind an AM radio dial provides a capacitance of 1nF when the dial setting is “80”. Calculate the inductance of the associated coil.
5. A series circuit comprises an inductor with resistance $10\ \Omega$ and inductance $159\ \mu\text{H}$, and a variable capacitor, connected to a 50-mV sinusoidal supply of frequency 1 MHz.
 - a. What value of capacitance will result in resonant conditions, and what will be the current at resonance?
 - b. For what values of capacitance will the current at this frequency be reduced to 10% of its value at resonance?

6. A generator supplies a variable-frequency voltage of constant amplitude = 10 V (rms) to the circuit of Fig below where $R = 5\ \Omega$, $L = 4\ \text{mH}$, and $C = 0.1\ \mu\text{F}$. The frequency is to be varied until a maximum rms current flows. Predict the maximum current, the frequency at which this occurs, and the resulting voltages across the inductance and capacitance.



7. A series circuit is to be resonant at 800 kHz. Specify the value of C required for a given inductor ($L = 40\ \mu\text{H}$ and $R = 4.02\ \Omega$) and predict the bandwidth. Assume that the capacitor is ideal, i.e., introduces no resistance.
8. A series-connected circuit has $R = 4\ \Omega$ and $L = 25\ \text{mH}$.
- (a) Calculate the value of C that will produce a quality factor of 50.
 - (b) Find ω_1 , ω_2 , and BW (bandwidth).
 - (c) Determine the average power dissipated at $\omega = \omega_r$, ω_1 , and ω_2 .
- Take Voltage source rms value as 100 V.
9. For an FM radio receiver, the incoming wave is in the frequency range from 88 to 108 MHz. The tuner circuit is a parallel RLC circuit with a $4\ \mu\text{H}$ coil. Calculate the range of the variable capacitor necessary to cover the entire band.